

Elastic Detection Portfolio – Combined Validation Summary

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Executive Overview

This portfolio section demonstrates the successful implementation and validation of two core Windows-based detection scenarios within Elastic Security:

1. **Suspicious PowerShell Flags (EQL)** – indicative of potential script-based attack execution.
2. **Failed Logon Burst (Event ID 4625)** – representing repeated authentication failures and potential brute-force attempts.

These detections collectively validate Elastic Security's capability to detect and respond to both execution-based and authentication-based attack surfaces using the Elastic Defend and System integrations. The artifacts produced confirm telemetry flow, rule execution, and alert generation across the Elastic Stack 9.2.0 environment.

Environment Summary

- **Platform:** Elastic Stack 9.2.0 (Kibana, Fleet, Elastic Agent)
- **Host OS:** Windows 11 (Endpoint with Elastic Agent 9.2.0)
- **Integrations:**
 - Elastic Defend (Endpoint telemetry, process events)
 - System Integration (Windows Security event collection)
- **Agent Status:** Enrolled and healthy in Fleet
- **Data Streams:** `logs-endpoint.events.\*`, `system.security`

Detection 1: Suspicious PowerShell Flags (EQL)

Objective

Detect potential malicious PowerShell usage characterized by suspicious flags such as `-enc`, `-nop`, or `-NoProfile`.

Methodology

1. **Telemetry Verification:** Validated ingestion of PowerShell process events through the Elastic Defend integration.
2. **Rule Creation:** Configured a custom EQL rule matching PowerShell command-line arguments containing known obfuscation or evasion flags.
3. **Rule Execution:** Triggered rule manually and verified matches in Discover using KQL queries referencing `kibana.alert.rule.name`.
4. **Alert Validation:** Confirmed detection of PowerShell executions with encoded or no-profile flags.

Result

Elastic Security correctly identified suspicious PowerShell command executions, producing alerts associated with the EQL rule "Suspicious PowerShell Flags (EQL)." This validates detection for T1059.001 (PowerShell) per the MITRE ATT&CK framework.

Key Artifacts

- 12_rule_manual_run.png – Manual rule execution screen
- 13_agent_reenrolled_healthy.png – Agent verification post-rule deployment
- 14_discover_verified_powershell.png – Process telemetry verified in Discover
- 15_discover_cmdline_flags.png – Command-line flags verified

- 16_rule_manual_run_success.png – Rule executed successfully

Detection 2: Failed Logon Burst (Event ID 4625)

Objective

Detect excessive failed Windows logon attempts (Event ID 4625), simulating a brute-force or credential-stuffing attack.

Methodology

1. ****Enable Security Log Collection:**** Configured the System integration to include the `security` channel for Windows Event Logs.
2. ****Verify Ingestion:**** Confirmed appearance of 4625 events in `system.security` using KQL (`winlog.event\_id: 4625 OR event.code: 4625`).
3. ****Detection Rule Creation:**** Implemented a threshold rule alerting on 5+ failed logons within 5 minutes.
4. ****Simulation:**** Generated repeated failed logons via Windows login interface and command-line `runas.exe` executions.
5. ****Validation:**** Verified alert generation in Elastic Security, confirming proper telemetry and rule execution.

Result

The rule successfully triggered on multiple failed logon attempts, validating both event ingestion and detection correlation between GUI and command-line sources.

Key Artifacts

- 19_system_integration_security_enabled.png – System integration configuration with Security logs enabled
- 20_discover_system_security_hits.png – Confirmation of 4625 events in Discover
- 21_rule_4625_threshold_definition.png – Custom detection rule configuration
- 22_4625_alerts_triggered.png – Alert generation following simulated brute-force attempts

Verification and Outcome

Both detections were validated through manual event simulation, telemetry confirmation, and alert generation. The tests demonstrate full end-to-end visibility—from data collection to detection logic execution—in Elastic Security.

Detection	Technique	Data Source	Result
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Suspicious PowerShell Flags	MITRE ATT&CK; T1059.001	Endpoint Process Events	■ Alerted successfully
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Failed Logon Burst (4625)	MITRE ATT&CK; T1110	Windows Security Logs	■ Alerted successfully
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Conclusion

These validated detections confirm the organization's ability to identify critical Windows-based attack behaviors using Elastic Security. The integration of both endpoint and system telemetry enables detection coverage for execution abuse and credential attacks—providing a foundation for future SOC

automation and threat-hunting initiatives.

****Artifacts 01–22**** collectively evidence the design, implementation, and validation of both detections. All playbooks, configurations, and results are reproducible for inclusion in professional and academic portfolio contexts.